Is Milnepan effective in reducing the risk of death among patients with heffpox?

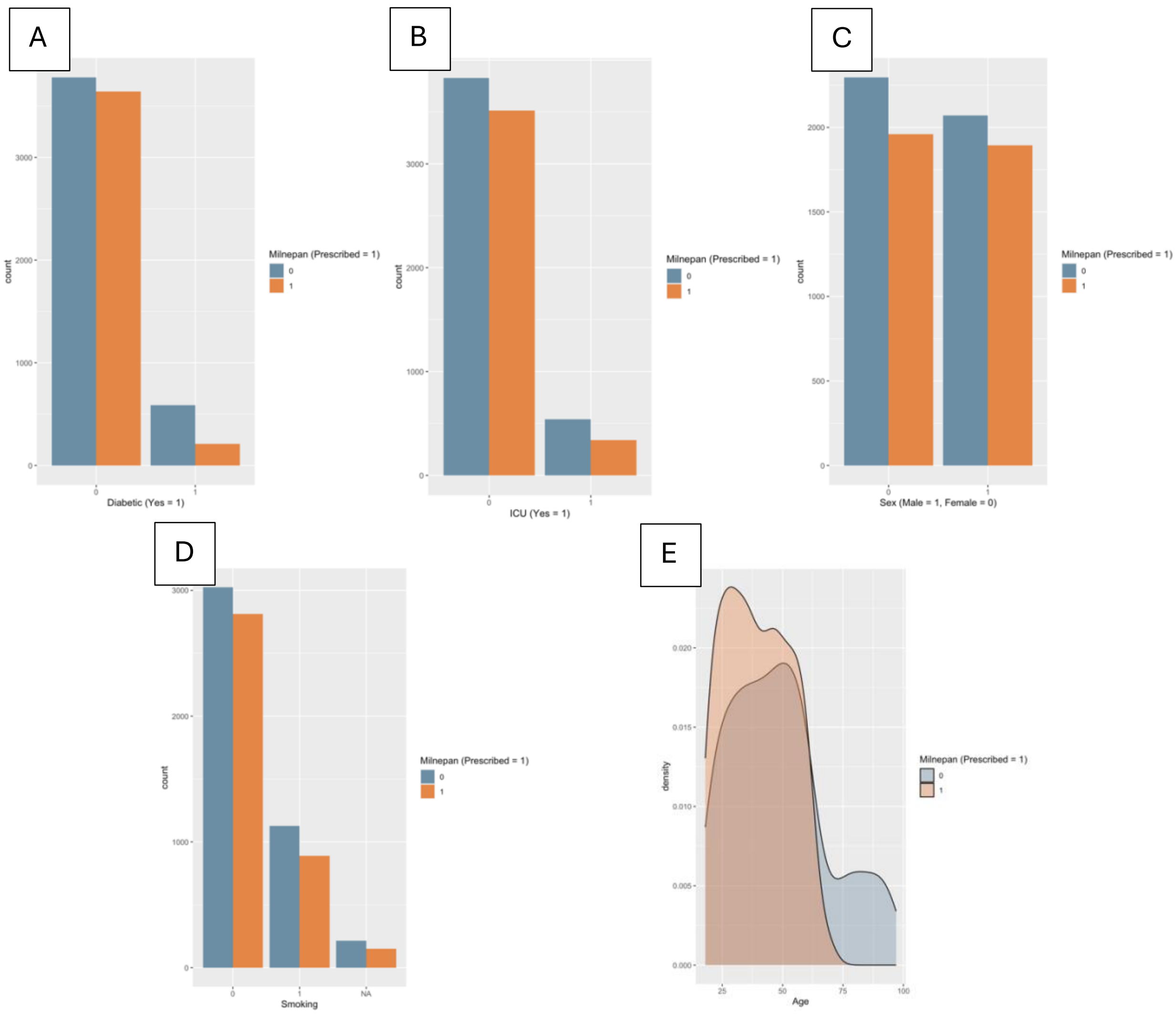
In this report we will be using the data from the hospital to investigate whether milnepan reduces the risk of death by day 7 among heffpox patients focussing on two methods to adjust for confounding- propensity score matching and regression adjustment.

1. Exploratory Data Analysis

For our analysis we were only interested in patients with heffpox, however the full original dataset contained some patients without heffpox. We therefore filtered the data to include only heffpox patients, our population of interest. This dataset contains data from 8219 heffpox patients. Below is a table summarising the contents of this filtered dataset.

Variables	Description and Key statistics
Heffpox	0= no heffpox, 1= heffpox. Max and min are both one- successfully filtered.
Sex	0= female, 1= male. 48.23% are male
Age	Age range 18-97. Median is 43 and mean is 48.3%
BMI	Range 13.61- 42.80. Mean is 27.97 and median is 27.99. There are 2718 missing values
Smoking	0= not current smoker, 1= current smoker. 25.68% of the patients were current smokers. 365 missing values
Diabetes	0= no diabetes, 1= diabetes. 9.72% of the patients had diabetes
Milnepan	0= no milnepan, 1= milnepan. 46.87% of the patients were given milnepan
ICU	0= no icu, 1= icu admission. 10.68% of patients were admitted to icu.
TEVENT	Time from admission until death or censoring. Range is 0.00034- 10. Median is 10 days. Mean is 6.84
Status	0= right censored, 1= died. 49.79 % died by the end of the data collection period
Death7day	40.60% died by day 7

Table 1- Summary of the variables contained in the dataset.



<u>Figure 1</u>- Bar plots and density plot showing distributions of A) diabetes, B) icu , C) Sex, D) Smoking and E) Age grouped by milnepan treatment.

<u>Analysis of EDA distribution graphs</u>

In figure 1, we can see the distributions of some the variables grouped by treatment. In figure 1A, we can see that majority of the heffpox patients in this study are not diabetic and that the proportion of diabetic patients that receive treatment is a lot less than those that do. This makes sense given we were told diabetes is shown to interfere with milnepan, so this was most likely why less diabetic patients were given this treatment. In figure 1B, we can see that majority of the patients were not admitted to the icu and that the proportion of treated and non-treated patients appears to be similar for icu and non-icu patients. In figure 1C, the proportion of male and female patients prescribes milnepan appears to be similar with slightly less males prescribed overall than females. In figure 1D, we can see that majority of the patients are not current smokers and that the the proportion of patients treated for both smoking groups appears to be similar. The smoking variable also has missing values. There is a slightly higher count of patients non treated with missing values than those that were treated. We will come back to handling the missing values in the next section. Finally, figure 1E shows that there is a difference in age distributions among treated and untreated groups. Patients that are treated with milnepan tend to be younger, with a peak at around 30 years. The untreated group peaks at around 50 years and contains more patients over the age of 60 than the treated group.

TEVENT Analysis

We have two variables giving us information about death- TEVENT and Death7day. Since majority of those that died did die by day 7 it would be best to focus on the Death7day for most of our analysis. However, we can still look at the TEVENT variable to give us an idea of the survival rates for heffpox patients as well as difference in survival between those that were treated with milnepan and those that were not. This of course would not account for any confounders which we will address later in our causal analysis with the Death7day variable.

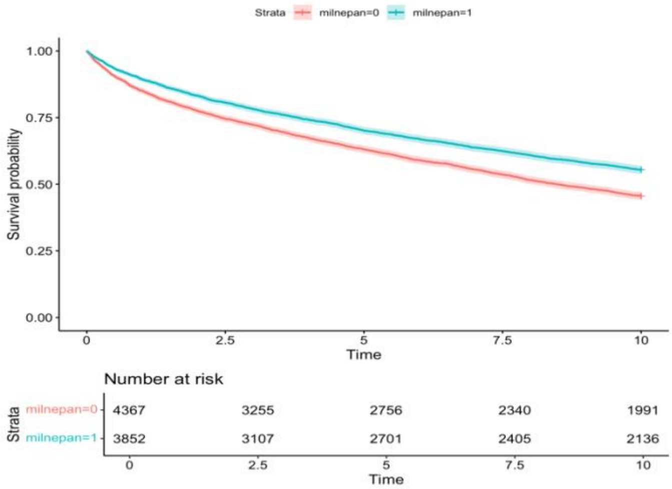


Figure 2- Kaplan Meier Curves with confidence intervals showing survival probability of heffpox patients given milnepan vs those that were not given it. Number at risk gives a count of patients that are still alive at specific time points for both treatment groups.

In the Kaplan Meier curves in figure 2, we can see that both curves are not the same- the median survival for untreated patients is around 10 days whereas for treated it is greater (censored patients mostly treated). The curve for no milnepan shows a greater decline in survival probability over time and the difference between the curves increases over time. The confidence intervals do not overlap either. We can formally test this difference using the log rank test.

Null Hypothesis: Both groups have identical survival curves.
Alternate Hypothesis: The curve for those given milnepan vs those that were not given milnepan are different.

Assumptions for the log rank test are the same as the assumption for the Kaplan Meier curves with a key assumption being non- informative censoring. In non-informative censoring we assume that the reason behind the censoring is not actually related to the outcome probability (which is death in this case). We have right-censored data so assuming our censoring is non-informative the reason behind the censoring is either that the patient was lost to follow up or still alive at the end of this study. Since the time is short (10 days) and our outcome is death, the latter is the reason that makes most sense for our data.

Doing the log rank test in R gives a chi-squared value of 83.4 on 1 degrees of freedom with a p-value of <2e-16. Therefore, at a 5% significance threshold we can reject the null hypothesis. The difference between these curves is significant.

2. Handling missing data

From our exploratory data analysis, we saw that we had missing values for the variables BMI and Smoking. The BMI column contained the most missing values with 33.07% missing values and smoking contained fewer missing values with 4.44% missing. To find the best approach for handling these missing values it is important to understand the mechanism behind its missingness. The easiest to handle is if the data is missing completely at random (MCAR) meaning that the probability of missingness is unrelated to any variables. Missing at random (MAR) is when missingness can be explained by other variables we have measured and missing not at random (MNAR) is when missing data depends on variables we do not have data for.

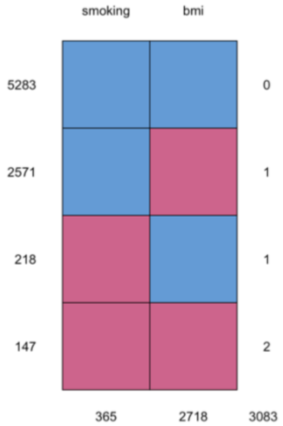


Figure 3- diagram showing missingness pattern between smoking and bmi

Missingness model (logistic regression)	Associated variables
Is missing for smoking	• Age • Sex • Diabetes
Is missing for bmi	• Age • Sex • Diabetes

Table 2- Table containing variables associated with missingness for smoking and for bmi. Determined using logistic regression in R

From the analysis results from figure 3 and table 2 it seems that missingness in both smoking and bmi are related to the variables age, sex and diabetes in our data therefore our data is likely missing at random (MAR). We can now proceed to handling these missing values. For this we have chosen to use multiple imputation which is considered the gold standard method for handling missing values. We can do this in R using the mice package. Here we will end up with multiple datasets each containing different imputed values for the missing data. Since we can never know what the actual values should have been it is good to have multiple possible values to account for that uncertainty. For our analysis we will create 10 imputed datasets using predictive mean matching (PMM) to get the values. The imputed values for each of them need to be checked to ensure that the values are plausible (distribution should be similar to original bmi and smoking) which they are for our data (see figure 4 and table 3 below).

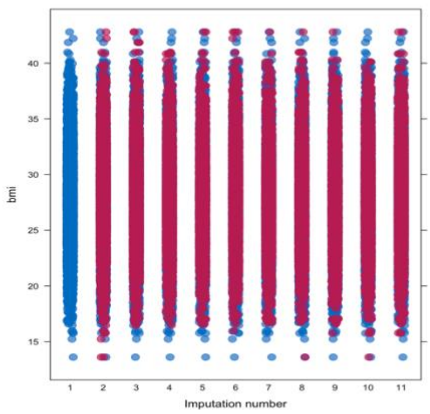


Figure 4- strip plot showing bmi distribution of original and imputed datasets

Original	25.68%
Imputation 1	25.75%
Imputation 2	25.67%
Imputation 3	25.91%
Imputation 4	25.75%
Imputation 5	25.84%
Imputation 6	25.83%
Imputation 7	25.72%
Imputation 8	25.96%
Imputation 9	25.84%
Imputation 10	25.81%

Table 3- Proportion of current smokers in original dataset and each of the ten imputed datasets.

3. Causal Analysis

We want to know whether taking milnepan would reduce the risk of death by day 7 amongst patients with heffpox. This type of question that deals with counterfactuals is usually best answered through conducting randomised controlled trials which is not always possible (for ethical reasons for example). Observed data would contain confounders which are variables related to both the exposure (milnepan treatment) and the outcome (death by day 7) that may falsely make it seem as though the exposure and outcome are related. In causal inference we aim to first identify what we believe could be potential confounders in our study using expert knowledge and then adjust for these confounders so that treatment groups are exchangeable. Exhangeability is important as it means we can be more certain that the relationship between milnepan treatment and death by day 7 exists and is not due to some confounder bias. Identifying confounders can be difficult though and often we use directed acyclic graphs (DAGs) to identify which confounders to adjust for. Below in figure 5 is a DAG for our analysis created prior to looking at the trends within the data (DAG is independent of our data). In our DAG we identified diabetes and age as potential confounders to adjust for.

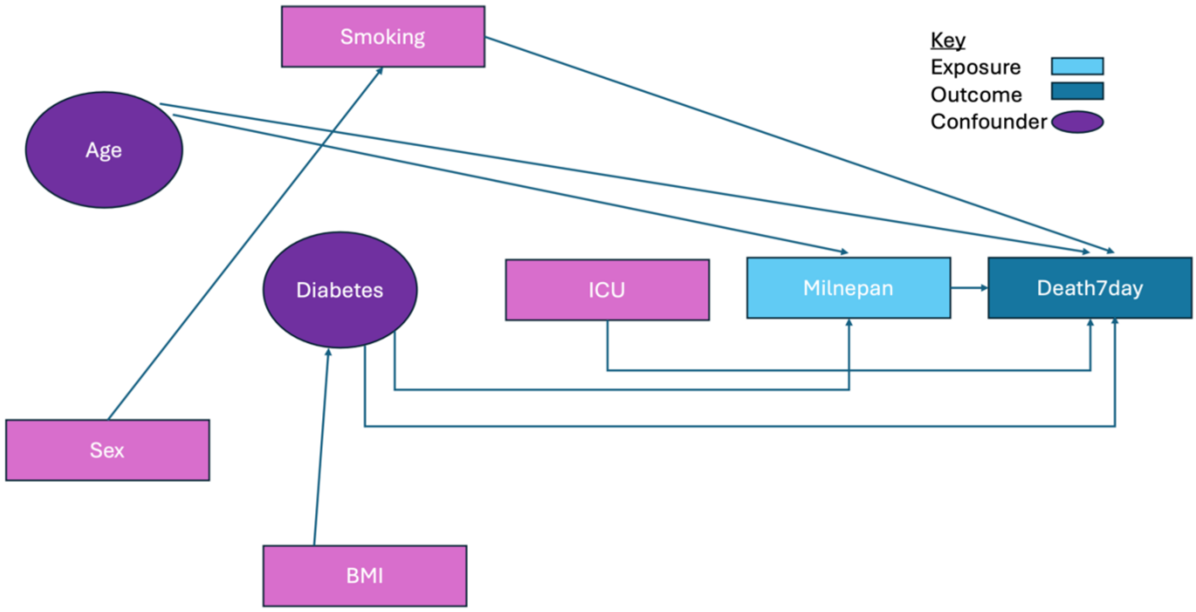


Figure 5- Directed acyclic graph to help visualise relationships between variables, particularly for identifying potential confounders for the relationship between milnepan and death by day 7.

Before moving on to the analysis, we should check whether the potential confounders identified from the DAG are actual potential confounders in our data. Since our outcome and exposure are both binary variables, we used logistic regression models to check this. The first model death7day ~ age + diabetes showed both age and diabetes to be significantly associated with death by day 7 at a 5% threshold with p-values of 2e-16 for both variables. The second model milnepan ~ age + diabetes showed that age and diabetes were also significantly associated with milnepan at 5% threshold also with p-values of 2e-16 for both variables. Since age and diabetes are both associated significantly with milnepan and death by day 7, they are likely to be confounders in our data. We need to adjust for them in our analysis.

There are many methods to adjust for confounders in causal analysis but the two that we will be focussing on and comparing our results for are propensity score matching and regression adjustment

Propensity Score Matching

We have already identified our potential confounders as age and diabetes, and we produced plots for the distributions for them both in figure 1 (1A and 1E). The distribution between treatment groups is different for these variables so we need to get propensity scores for each observation for each data and include our confounders in this, so we have a better balance of age and diabetes for both groups. Since we have multiple imputed datasets, instead of using matchit package for getting propensity scores and combining them to our datasets we will use the matchthem package in r which is designed to do this for multiple imputation data sets. The documentation for this package written by Pishgar et.al. (2024) can be found in the reference section.

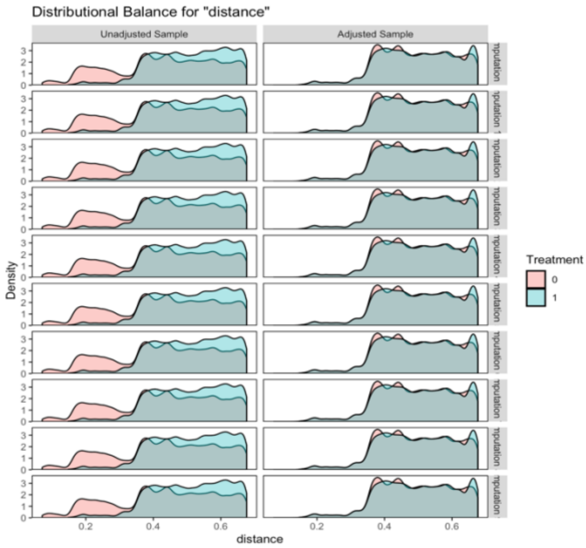


Figure 6- Distributional balance of propensity scores ("distance") before and after adjustment for each of the 10 imputed datasets

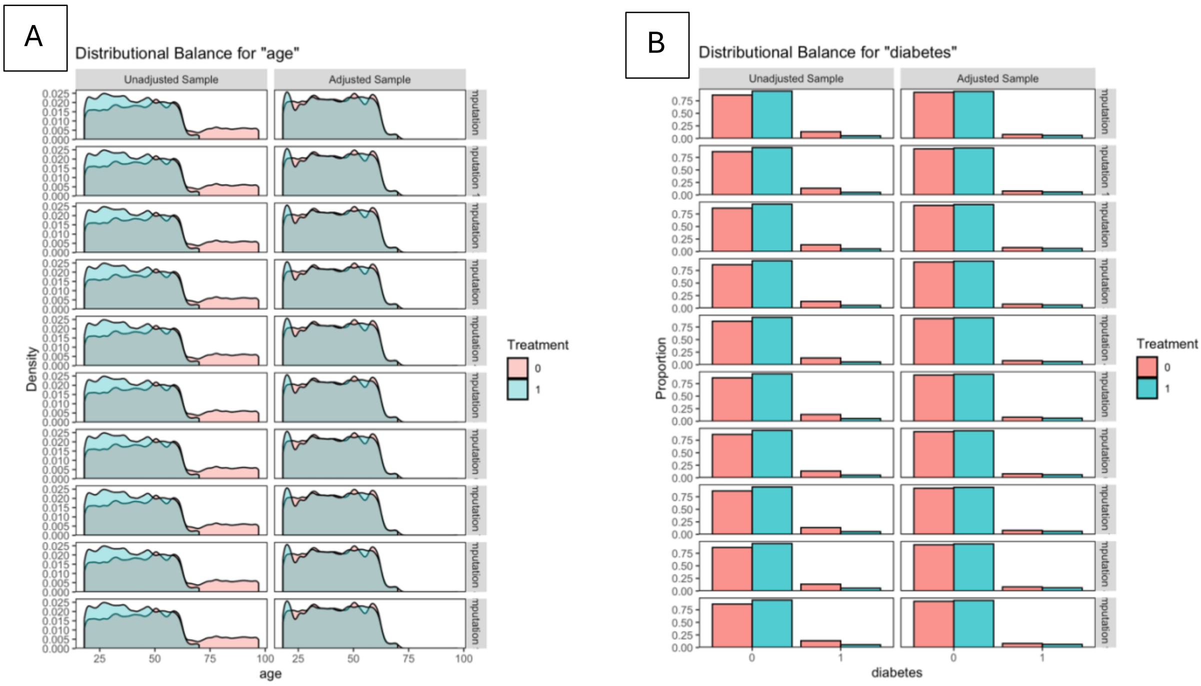


Figure 7- Distributional balance for A) age B) diabetes before and after adjustment for each of the 10 imputed datasets.

In figure 6, allows us to check the positivity assumption has been met sufficiently and to see the balance of propensity scores after matches are created. We see that there is enough overlap in the distributions of our propensity scores across both treatment groups before the matches so is sufficient for assumption of positivity. After the matching, the balance of propensity scores is much better- the distributions of the treatment groups overlap a lot more than in the unadjusted sample.

To ensure that we have achieved the balance between our confounders age and diabetes between both treatment groups, we must look at the distributions of these confounders before and after adjustment. Figure 7 shows improvement in the balance of age (7A) and diabetes (7B) between the treatment groups after propensity score matching. Exchangeability should now have been achieved between groups.

Now that the potential confounding has been adjusted, we can carry out the causal analysis. Since we have a binary outcome variable (death7day), we used logistic regression to carry out our analysis and with only our exposure variable milnepan as a covariate in the model. Since we worked with imputed datasets, we used the with() function to carry out this analysis on each of our matched and imputed datasets and then pooled together these results to get the final estimate. After exponentiating our log odds estimate, we get odds ratio of 0.89 as our coefficient. With a p-value of 1.13 ($1.38e^{-02}$) this estimate was not significant at a 5% significance threshold.

Regression adjustment

In regression adjustment we include the potential confounders as covariates in our logistic regression model interpreting only the coefficient for our exposure variable milnepan. This is because the milnepan estimate would give us the log odds of death by day 7 for those who take milnepan when the other covariates in the model are held constant. This model in R gives us log odds of -0.072, which after exponentiating gives us odds ratio of 0.93. This estimate was not significant at a 5% threshold with a p-value of 1.22 ($1.35e^{-01}$)

Comparison between regression adjustment, propensity score matching and unadjusted model

Model	**Milnepan estimate**	**Std.error**	**Statistic**	**Df**	**P-value**
Pooled model for no adjustment for potential confounders	-0.34 (odds ratio = 0.71)	0.045	-7.57	8214. 10	$4.14e^{-14}$ (significant at 0.05)
Pooled model after Regression adjustment	-0.072 (odds ratio= 0.93)	0.048	-1.50	8212.10	$1.35e^{-01}$ (not significant)
Pooled model after propensity score matching	-0.12 (odds ratio = 0.89)	0.050	-2.46	6803.27	$1.38e^{-0.2}$ (not significant)

Table 4- Table showing outputs for milnepan estimate for three different models- no adjustment, regression adjustment and propensity score matched.

4. Conclusion

In table 4 we can see that the odds ratio for all three models shows odds ratio less than 1 meaning that treatment with milnepan decreases the odds of death. However, the adjusted models found this decrease to be non-significant. So based on our analysis, there is no causal relationship between treatment with milnepan and death by day 7. The effect of treatment shown in the non-adjusted model was likely a result of confounding bias from age and diabetes. Both adjustment methods worked here to remove the confounding bias although the models for both methods gave slightly different odds ratios. Odds ratio was higher for the regression adjustment model which showed odds ratio of 0.93 whereas for propensity score matching it was 0.89.

5. Limitations of our analysis

It is important to note that our analysis did have some limitations. Firstly, for the missing values we imputed, we can never be certain as to what those missing values would be although we did our best to account for uncertainty by using multiple imputation. We also concluded that the missing data was MAR, however it is a possibility that the data was in fact MNAR since it is possible that a metric such as bmi is influenced by people that are reluctant to share theirs or perhaps bmi was only taken for people with comorbidities not observed in our data. This would result in potentially biased imputed values.

Another limitation in our analysis is that we cannot be certain that we selected the correct confounders for our model. We depended on the DAG we drew based on our own knowledge and information given to us in the brief, so we strongly assumed that it was correctly drawn. If this DAG was drawn incorrectly, the confounders we picked for our analysis would be incorrect. Having more knowledge on the area of interest would help us draw a more accurate DAG.

6. Reference:

Pishgar, F. et al. (2024). MatchThem: Matching and Weighting Multiply Imputed Datasets. Available from: https://cran.r-project.org/web/packages/MatchThem/ [Accessed February 1, 2025].

7. Appendix

```
library(tidyverse)
heffpox <- read_rds("Documents/heffpox_2425.RData")
# Loading more required packages
library(mice)
library(survival)
library(survminer)
library(ggplot2)
head(heffpox)
view(heffpox)
sum(is.na(heffpox)) # has 3315 missing values
summary(heffpox)
filtered_df <- heffpox %>% filter(heffpox == 1)
head(filtered_df)
view(filtered_df)
summary(filtered_df)
TEVENT_fit <- survfit(Surv(filtered_df$TEVENT, filtered_df$Status)~ 1, conf.type= "log-log")
summary(TEVENT_fit)
ggsurvplot(TEVENT_fit, data= filtered_df, conf.int= TRUE, risk.table= TRUE)
KM_heffpox <- survfit(Surv(filtered_df$TEVENT, filtered_df$Status)~ filtered_df$milnepan,
conf.type= "log-log")
ggsurvplot(KM_heffpox, data= filtered_df, conf.int= TRUE, risk.table= TRUE)
survdiff(Surv(TEVENT, Status)~ milnepan, data= filtered_df)
table(filtered_df$milnepan, filtered_df$death7day)
treat_death <- ggplot(filtered_df, aes(x=as.factor(milnepan),
                    fill= as.factor(death7day))) +
  geom_bar(position = "dodge") +
  xlab("Milnepan (Prescribed = 1)") +
  labs(fill = "Death7day (Yes = 1)") +
  scale_fill_manual(values = c("#6b8ea4", "#cccccc"))

diabetes_milnepan <-  ggplot(filtered_df, aes(x=as.factor(diabetes)  fill= as.factor(milnepan))) +
```

```
  geom_bar(position = "dodge") +
  xlab("Diabetic (Yes = 1)") +
  labs(fill = "Milnepan (Prescribed = 1)") +
  scale_fill_manual(values = c("#6b8ea4", "#e48646"))
diabetes_milnepan
sex_milnepan <- ggplot(filtered_df, aes(x= as.factor(sex),
                                        fill = as.factor(milnepan))) +
  geom_bar(position = "dodge")  +
  xlab("Sex (Male = 1, Female = 0)") +
  labs(fill = "Milnepan (Prescribed = 1)") +
  scale_fill_manual(values = c("#6b8ea4", "#e48646"))
sex_milnepan
# Among Smokers, how many were given milnepan? Also some counts for NA values.
smoking_milnepan <- ggplot(filtered_df, aes(x= as.factor(smoking),
                                        fill = as.factor(milnepan))) +
  geom_bar(position = "dodge")  +
  xlab("Smoking") +
  labs(fill = "Milnepan (Prescribed = 1)") +
  scale_fill_manual(values = c("#6b8ea4", "#e48646"))
smoking_milnepan
library("hrbrthemes")
age_milnepan <- ggplot(filtered_df, aes(x= age,
                                     group = as.factor(milnepan),
                                     fill = as.factor(milnepan))) +
  geom_density(adjust = 1.5, alpha =0.4)  +
  xlab("Age") +
  labs(fill = "Milnepan (Prescribed = 1)") +
  scale_fill_manual(values = c("#6b8ea4", "#e48646"))
icu_milnepan <- ggplot(filtered_df, aes(x= as.factor(icu),
                                     fill = as.factor(milnepan))) +
  geom_bar(position = "dodge")+
  xlab("ICU (Yes = 1)") +
  labs(fill = "Milnepan (Prescribed = 1)") +
  scale_fill_manual(values = c("#6b8ea4","#e48646"))

figure1 <- ggarrange(diabetes_milnepan, sex_milnepan, smoking_milnepan,
                    icu_milnepan, age_milnepan,
                    labels = c("A", "B", "C", "D", "E"),
                    ncol = 3, nrow = 2)
# Handling missing data
# Quantifying the missingness
colMeans(is.na(filtered_df))*100 # 33% of the data missing for BMI
# Looking further at the pattern of missingness
table(is.na(filtered_df$smoking),is.na(filtered_df$bmi))
md.pattern(filtered_df[, c("smoking", "bmi")])

# missing smoking
glm_missing_smoking <- glm(is.na(smoking) ~ age + sex + diabetes + icu, family= "binomial",data =
filtered_df)
summary(glm_missing_smoking)
#missing bmi
glm_missing_bmi <- glm(is.na(bmi) ~ age + sex + diabetes + icu, family= "binomial",data = filtered_df)
summary(glm_missing_bmi)
# imputing the missing values- imputation model
data_imputed <- mice(filtered_df, seed= 1, m=10, method = "pmm")
# diagnostics to check plausibility of imputation- for bmi
```

```
stripplot(data_imputed,bmi ~ .imp, pch = 20, cex = 2)

all_imputed <- complete(data_imputed, action= "all")
smoking_mean1 <- mean(imputed_1$smoking)
smoking_mean2 <- mean(imputed_2$smoking)
smoking_mean3 <- mean(imputed_3$smoking)
smoking_mean4 <- mean(imputed_4$smoking)
smoking_mean5 <- mean(imputed_5$smoking)
smoking_mean6 <- mean(imputed_6$smoking)
smoking_mean7 <- mean(imputed_7$smoking)
smoking_mean8 <- mean(imputed_8$smoking)
smoking_mean9 <- mean(imputed_9$smoking)
smoking_mean10 <- mean(imputed_10$smoking)
age_milnepan <- ggplot(filtered_df, aes(x=age, color=as.factor(milnepan)))+ geom_density()
summary(glm(milnepan ~ age + diabetes,
        family = "binomial", data = imputed_1))
summary(glm(death7day ~ age + diabetes,
        family = "binomial", data = imputed_1))
# Model without accounting for confounders- for comparison
no_confounders <- with(data_imputed, glm(death7day ~ milnepan, family = "binomial"))
pooled_no_confounders <- pool(no_confounders)
summary(pooled_no_confounders)
model1 <- with(data_imputed, glm(death7day ~ milnepan + diabetes + age, family = "binomial"))
pooled_regression <- pool(model1)
### adjusting for confounders method 2- Propensity Score Matching
library(MatchThem)
library(cobalt)
psm <- matchthem(milnepan ~ diabetes + age, data= data_imputed, method= "nearest",
          caliper = 0.1)
summary(psm)
imp_with_ps <- complete(psm, action = "all")
bal.tab(psm, m.threshold = 0.1)
bal.plot(psm, which= "both")
bal.plot(psm, "age", which= "both")
bal.plot(psm, "diabetes", which= "both")
model2 <- with(psm, glm(death7day ~ milnepan, family = "binomial"))
pooled_psm <- MatchThem::pool(model2)
summary(pooled_psm)
```